



Computing on Masked Data for Privacy Protection

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Agenda

- Project Overview
- Goals
- Current Progress
- Timeline
- Next Steps



Project Overview

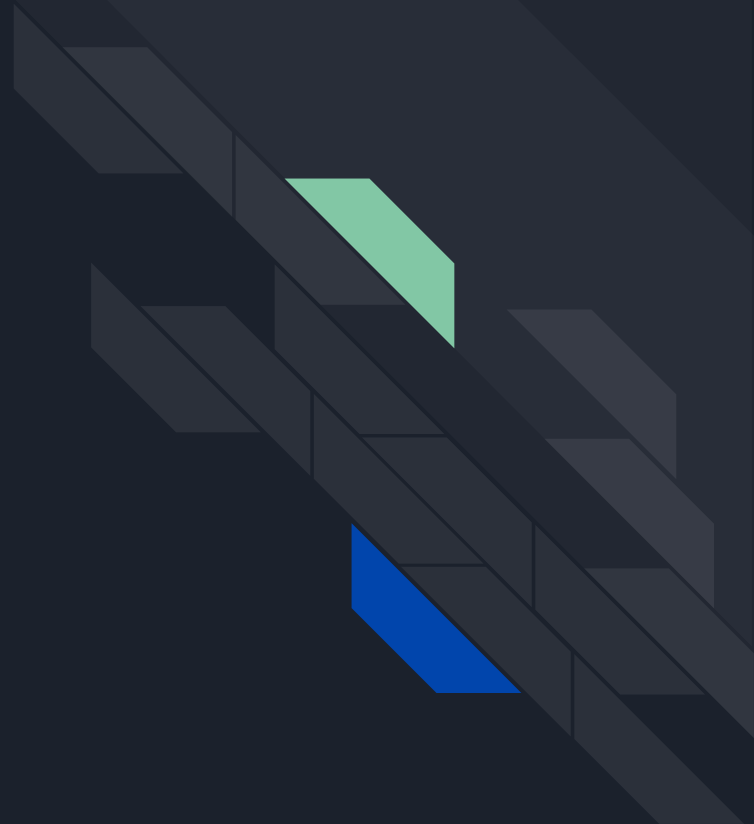
- With privacy preservation becoming an increasingly challenging problem, many models of masking data have been introduced.
- The main goal of this project is to identify approaches and designs for masking data to preserve privacy.



Goals

- Analyze existing data masking models
- Suggest features that increase the privacy protections for different types of system architectures

Data Masking





Types of Data Masking

- Static data masking
 - Alters data at rest
 - Permanent
- Dynamic data masking
 - Alters data in transit
 - Does not affect original data



Encryption

- Uses a key to encode text
- Advantages
 - High cost to attack
 - Low disclosure risk
- Disadvantages
 - High cost to implement
 - Excessive in most cases



Substitution

- Replace the existing data in the database with random values from a pre-prepared dataset
- Advantages
 - Low disclosure risk
- Disadvantages
 - The pre-prepared dataset must be available
 - Low data utility

Original Data	Masked Data
Aaric	Aaron
Adam	Aaron



Shuffling

- Swap data fields across different records
- Advantages
 - Good for computing on whole dataset
- Disadvantages
 - High disclosure risk
 - Low data utility when not using whole dataset



Nulling Out

- Replace the sensitive data with NULL values
 - Phone number -> (XXX)XXX-XXXX
 - Social Security Number (SSN) -> XXX-XX-XXXX
- Advantages
 - No disclosure risk
- Disadvantages
 - No data utility



Masking Out

- Extension of nulling out
- Advantage:
 - Higher data utility
- Disadvantage:
 - High disclosure risk

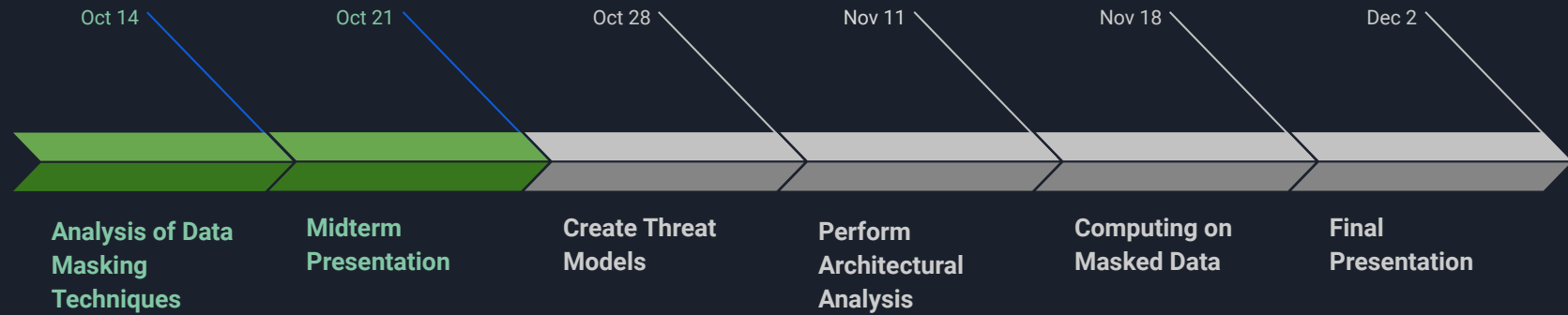
Original	4297 8296 7496 8724
Too Little	4297 8296 7496 87XX
Too Much	XXXX XXXX XXXX XX24
Just Enough	4297 XXXX XXXX 8724

Policy-Based Data Masking

- California Consumer Privacy Act (CCPA)
- Healthcare Insurance Portability and Accountability Act (HIPAA)
- General Data Protection Regulation (GDPR)
- The Gramm-Leach-Bliley Act (GLBA)



Timeline





Next Steps

- What is the most efficient AND effective data masking solution?
- How do malicious cyber actors use data masking to defeat cybersecurity?
- Is a hybrid data masking technique feasible?
- Is a hybrid masking architecture feasible?
- Determine viability of masking techniques across different architectures

Questions

